DBMS Lab Assignment 2 - Solution

DDL and DML on EMP1 and EMP_trainee: ALTER, RENAME, INSERT, UPDATE, DROP COLUMN and TRUNCATE. Each step shows the Oracle SQL followed by its output.

*Note: outputs were produced by running equivalent statements on a SQLite database, formatted like SQL*Plus. The DESC listings show the Oracle column types.*

1. Create table EMP1

SQL

```
CREATE TABLE EMP1 (
  ID          NUMBER(2),
  Name        VARCHAR2(10),
  Basic       NUMBER(6,2),
  Designation VARCHAR2(10),
  Age         NUMBER(2));

DESC EMP1;
```

OUTPUT

```
Table created.

SQL> DESC EMP1
Name         Null?  Type
-----------  -----  ------------
ID                  NUMBER(2)
NAME                VARCHAR2(10)
BASIC               NUMBER(6,2)
DESIGNATION         VARCHAR2(10)
AGE                 NUMBER(2)
```

2. Change Basic from float to integer

SQL

```
ALTER TABLE EMP1 MODIFY (Basic NUMBER(6));
DESC EMP1;
```

OUTPUT

```
Table altered.

SQL> DESC EMP1
Name         Null?  Type
-----------  -----  ------------
ID                  NUMBER(2)
NAME                VARCHAR2(10)
BASIC               NUMBER(6)
DESIGNATION         VARCHAR2(10)
AGE                 NUMBER(2)
```

Note: Basic NUMBER(6,2) has 4 digits before the decimal point, so NUMBER(6) keeps the same range as a whole number. Oracle only lets the precision/scale shrink while the column is empty, which is why this is done before inserting data.

3. Change Name size from 10 to 15

SQL

```
ALTER TABLE EMP1 MODIFY (Name VARCHAR2(15));
DESC EMP1;
```

OUTPUT

```
Table altered.

SQL> DESC EMP1
Name         Null?  Type
-----------  -----  ------------
ID                  NUMBER(2)
NAME                VARCHAR2(15)
BASIC               NUMBER(6)
DESIGNATION         VARCHAR2(10)
AGE                 NUMBER(2)
```

4. Create EMP_trainee with the same structure; rename ID to Emp_id

SQL

```
CREATE TABLE EMP_trainee AS SELECT * FROM EMP1 WHERE 1 = 2;
ALTER TABLE EMP_trainee RENAME COLUMN ID TO Emp_id;
DESC EMP_trainee;
```

OUTPUT

```
Table created.

Table altered.

SQL> DESC EMP_trainee
Name         Null?  Type
-----------  -----  ------------
EMP_ID              NUMBER(2)
NAME                VARCHAR2(15)
BASIC               NUMBER(6)
DESIGNATION         VARCHAR2(10)
AGE                 NUMBER(2)
```

Note: WHERE 1 = 2 copies the structure of EMP1 (with its changed sizes) but no rows.

5. Insert the data into EMP1

SQL

```
INSERT INTO EMP1 VALUES (1,'Rohit',6700,'Manager',24);
INSERT INTO EMP1 VALUES (2,'Sunil',6200,'Engineer',27);
INSERT INTO EMP1 VALUES (3,'Payal',6300,'Engineer',25);
INSERT INTO EMP1 VALUES (4,'Kunal',6700,'Trainee',28);
INSERT INTO EMP1 VALUES (5,'Sunita',6230,'Trainee',26);
INSERT INTO EMP1 VALUES (6,'Bimal',7000,'Trainee',25);
COMMIT;

SELECT * FROM EMP1;
```

OUTPUT

```
6 rows created.

ID  NAME    BASIC  DESIGNATION  AGE
--  ------  -----  -----------  ---
 1  Rohit    6700  Manager       24
 2  Sunil    6200  Engineer      27
 3  Payal    6300  Engineer      25
 4  Kunal    6700  Trainee       28
 5  Sunita   6230  Trainee       26
 6  Bimal    7000  Trainee       25

6 rows selected.
```

6. Insert all 'trainee' rows from EMP1 into EMP_trainee

SQL

```
INSERT INTO EMP_trainee
SELECT * FROM EMP1
WHERE  LOWER(Designation) = 'trainee';
COMMIT;

SELECT * FROM EMP_trainee;
```

OUTPUT

```
3 rows created.

EMP_ID  NAME    BASIC  DESIGNATION  AGE
------  ------  -----  -----------  ---
     4  Kunal    6700  Trainee       28
     5  Sunita   6230  Trainee       26
     6  Bimal    7000  Trainee       25

3 rows selected.
```

Note: LOWER() makes the match work for 'Trainee' as stored.

7. Add Skills and DOJ columns and fill them in

SQL

```
ALTER TABLE EMP1 ADD (Skills VARCHAR2(10), DOJ DATE);

UPDATE EMP1 SET Skills = 'Leadership', DOJ = DATE '2019-03-15' WHERE ID = 1;
UPDATE EMP1 SET Skills = 'Java', DOJ = DATE '2020-07-01' WHERE ID = 2;
UPDATE EMP1 SET Skills = 'Python', DOJ = DATE '2021-01-10' WHERE ID = 3;
UPDATE EMP1 SET Skills = 'SQL', DOJ = DATE '2024-06-03' WHERE ID = 4;
UPDATE EMP1 SET Skills = 'Testing', DOJ = DATE '2025-02-17' WHERE ID = 5;
UPDATE EMP1 SET Skills = 'Networking', DOJ = DATE '2025-08-11' WHERE ID = 6;
COMMIT;

DESC EMP1;
SELECT * FROM EMP1;
```

OUTPUT

```
Table altered.

1 row updated. (x6)

SQL> DESC EMP1
Name          Null?  Type
------------  -----  ------------
ID                   NUMBER(2)
NAME                 VARCHAR2(15)
BASIC                NUMBER(6)
DESIGNATION          VARCHAR2(10)
AGE                  NUMBER(2)
SKILLS               VARCHAR2(10)
DOJ                  DATE

ID  NAME    BASIC  DESIGNATION  AGE  SKILLS      DOJ
--  ------  -----  -----------  ---  ----------  ---------
 1  Rohit    6700  Manager       24  Leadership  15-MAR-19
 2  Sunil    6200  Engineer      27  Java        01-JUL-20
 3  Payal    6300  Engineer      25  Python      10-JAN-21
 4  Kunal    6700  Trainee       28  SQL         03-JUN-24
 5  Sunita   6230  Trainee       26  Testing     17-FEB-25
 6  Bimal    7000  Trainee       25  Networking  11-AUG-25

6 rows selected.
```

Note: Skills and joining dates are my own choice.

8. Update more than one row in one query

SQL

```
UPDATE EMP1
SET    Basic = Basic + 300
WHERE  Designation = 'Engineer';
COMMIT;

SELECT * FROM EMP1;
```

OUTPUT

```
2 rows updated.

ID  NAME    BASIC  DESIGNATION  AGE  SKILLS      DOJ
--  ------  -----  -----------  ---  ----------  ---------
 1  Rohit    6700  Manager       24  Leadership  15-MAR-19
 2  Sunil    6500  Engineer      27  Java        01-JUL-20
 3  Payal    6600  Engineer      25  Python      10-JAN-21
 4  Kunal    6700  Trainee       28  SQL         03-JUN-24
 5  Sunita   6230  Trainee       26  Testing     17-FEB-25
 6  Bimal    7000  Trainee       25  Networking  11-AUG-25

6 rows selected.
```

Note: One statement raised the Basic of both engineers (Sunil and Payal) by 300.

9. Rename Age to Age_in_Years in EMP1

SQL

```
ALTER TABLE EMP1 RENAME COLUMN Age TO Age_in_Years;
DESC EMP1;
```

OUTPUT

```
Table altered.

SQL> DESC EMP1
Name          Null?  Type
------------  -----  ------------
ID                   NUMBER(2)
NAME                 VARCHAR2(15)
BASIC                NUMBER(6)
DESIGNATION          VARCHAR2(10)
AGE_IN_YEARS         NUMBER(2)
SKILLS               VARCHAR2(10)
DOJ                  DATE
```

10. Drop the Age column from EMP_trainee

SQL

```
ALTER TABLE EMP_trainee DROP COLUMN Age;
DESC EMP_trainee;
SELECT * FROM EMP_trainee;
```

OUTPUT

```
Table altered.

SQL> DESC EMP_trainee
Name         Null?  Type
-----------  -----  ------------
EMP_ID               NUMBER(2)
NAME                 VARCHAR2(15)
BASIC                NUMBER(6)
DESIGNATION          VARCHAR2(10)

EMP_ID  NAME    BASIC  DESIGNATION
------  ------  -----  -----------
     4  Kunal    6700  Trainee
     5  Sunita   6230  Trainee
     6  Bimal    7000  Trainee

3 rows selected.
```

11. Rename the table to EMP_Mgr_Engr

SQL

```
RENAME EMP1 TO EMP_Mgr_Engr;
-- or: ALTER TABLE EMP1 RENAME TO EMP_Mgr_Engr;

SELECT * FROM EMP_Mgr_Engr;
```

OUTPUT

```
Table renamed.

ID  NAME    BASIC  DESIGNATION  AGE_IN_YEARS  SKILLS      DOJ
--  ------  -----  -----------  ------------  ----------  ---------
 1  Rohit    6700  Manager                24  Leadership  15-MAR-19
 2  Sunil    6500  Engineer               27  Java        01-JUL-20
 3  Payal    6600  Engineer               25  Python      10-JAN-21
 4  Kunal    6700  Trainee                28  SQL         03-JUN-24
 5  Sunita   6230  Trainee                26  Testing     17-FEB-25
 6  Bimal    7000  Trainee                25  Networking  11-AUG-25

6 rows selected.
```

Note: The sheet says 'rename EMP', but the table created is EMP1, so EMP1 is the one renamed.

12. Truncate EMP_Mgr_Engr

SQL

```
SELECT COUNT(*) AS rows_before FROM EMP_Mgr_Engr;
TRUNCATE TABLE EMP_Mgr_Engr;
SELECT COUNT(*) AS rows_after FROM EMP_Mgr_Engr;
DESC EMP_Mgr_Engr;
```

OUTPUT

```
ROWS_BEFORE
-----------
          6

1 row selected.

Table truncated.

ROWS_AFTER
----------
         0

1 row selected.

SQL> DESC EMP_Mgr_Engr
Name          Null?  Type
------------  -----  ------------
ID                   NUMBER(2)
NAME                 VARCHAR2(15)
BASIC                NUMBER(6)
DESIGNATION          VARCHAR2(10)
AGE_IN_YEARS         NUMBER(2)
SKILLS               VARCHAR2(10)
DOJ                  DATE
```

Note: TRUNCATE removes every row but keeps the table structure. It cannot be rolled back, unlike DELETE.